

Where human-video robot learning breaks when every hand wears a glove

Pipelines like [Phantom](#) (CoRL 2025) learn robot policies from human video by reading the operator's hand with a 3D hand-pose model, then turning the thumb and index fingertip into the gripper action. [Work gloves damage exactly that signal](#). I measured how much, and which capture choices win it back.

Code and detailed findings: github.com/adithyab94/glove-gap



Bare hand

Thin nitrile

Grey knit

Dark insulated

Insulated, bulkier

Bright insulated

Same recording, gloves painted onto real lab video where the true 3D hand position is known. That lets me measure error exactly.

3×

more finger error with a dark insulated glove than with a bare hand

16%

of real gloved hands found by the hand detector at matched size, against 100% of bare hands

12 cm

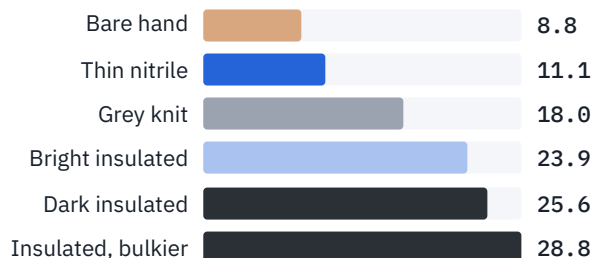
how far a single camera misplaces a hand in a bulky glove

FINDING 1

Gloves blur the fingers, and the fingers are the action

The pose model still sees the hand, but it can no longer read the finger shape. Grey knit doubles finger error and dark insulated gloves triple it. For a policy that learns gripper opening from fingertip distance, that is label noise in the one number that matters.

Finger-pose error, mm (lower is better)



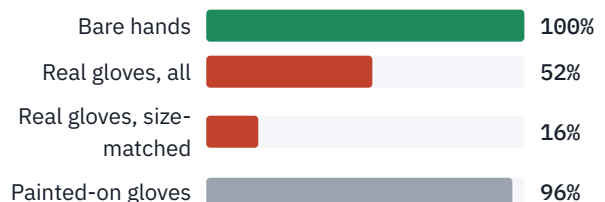
WiLoR, 19 held-out grasp sequences. HaMeR, the model Phantom uses, shows the same pattern.

FINDING 2

On real gloves, the hand often isn't found at all

On real photos of worn gloves, the detector misses about half of them. Small and mid-sized gloved hands, the typical view from a mounted camera, are almost never found. Painted-on gloves did not show this, so synthetic data alone would have hidden the biggest failure.

Share of hands found (higher is better)



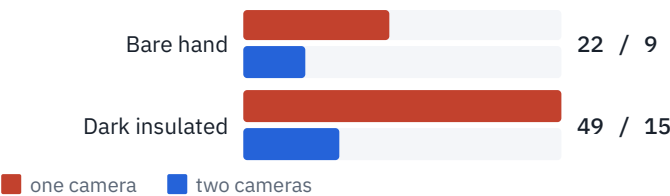
Real gloved hands the detector missed (public Roboflow photo set).

FINDING 3

More cameras fix where the hand is, not what the fingers do

One camera misjudges distance with gloves: a bulkier hand looks closer. Two calibrated cameras about 1 m apart remove most of that position error. They remove at most half of the finger error, so cameras alone don't rescue the gripper signal.

Hand position error, mm (lower is better)

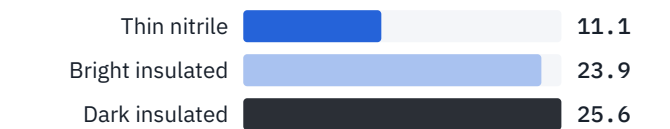


FINDING 4

Glove thickness matters. Glove colour doesn't.

A bright blue insulated glove is barely better than a dark one. The same blue in a thin nitrile glove keeps the finger detail. So the spec should name a thin, form-fitting glove; colour is a nice-to-have.

Finger-pose error, mm, same blue colour

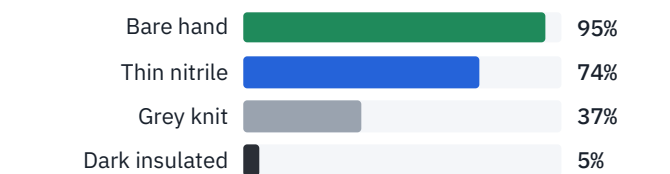


FINDING 5

A simple check flags bad batches before anyone trains on them

For each recorded episode I checked whether the two camera views agree on the hand. With bare hands almost every episode passes; with insulated gloves almost none do. It works as an alarm for a whole capture batch. It does not reliably rank single episodes.

Episodes passing the check (higher is better)



WHAT I'D PUT IN THE CAPTURE SPEC

ITEM	RECOMMENDATION	WHY
Hand model	A modern 3D hand model (HaMeR / WiLoR class), with its detector tested on our own gloves	The lightweight MediaPipe model finds 7% of real gloved hands
Gloves	Thin, form-fitting gloves wherever the task allows	Keeps finger error close to bare hands
Cameras	At least two calibrated views about 1 m apart, or a depth camera	Cuts position error from 49 to 15 mm with gloves
Batch check	Flag any capture batch where fewer than ~75% of episodes pass	Catches glove or setup problems before training money is spent

How I tested

- 19 held-out grasp recordings from DexYCB, a public lab dataset with true 3D hand positions
- Five glove types painted onto the hand, same frames each time
- Three hand-pose models, one and two cameras
- Real-glove check on 83 public photos

What this doesn't show yet

- Painted gloves, one person, lab objects
- No cold halls: fogging, low light
- The models may have seen this person in training, so trust the drops, not the absolute numbers

What I would test next

- Record a short session with real work gloves on a two-camera setup
- Measure how often the hand detector finds real gloved hands, and fine-tune it on gloved images if needed
- Check whether the batch alarm agrees with real pick logs